

A Comparative Study of Text-Classification Paradigms on 20 Newsgroups

Dan Waititu Wanjohi

Hochschule Luzern (HSLU) — NALAPRO Module

Email: danwwaititu@gmail.com

GitHub link: https://github.com/DanYT2/NLP_Project

WandB link: <https://wandb.ai/danwwaititu-hochschule-luzern/hslu-nalapro>

Abstract—This report studies how representation and adaptation strategy govern text-classification quality on the 20 Newsgroups corpus, progressing across four modelling paradigms of increasing capacity and a parameter-efficient bonus experiment. A fixed two-layer perceptron is first fed three classical representations: word2vec mean-pooling, TF-IDF, and a self-designed $\text{mean} \oplus \text{max}$ pooling variant. We then fine-tune bert-base-uncased with a hyperparameter sweep, add a stage of in-domain masked-language-model pretraining before fine-tuning, and finally probe a frozen Llama-3.2-3B under zero- and few-shot prompting. As a bonus, we QLoRA-fine-tune the same Llama model for sequence classification. On the full 7,532-document test set, QLoRA attains the best macro-F1 (0.748), ahead of a tuned BERT (0.704) and the strongest classical baseline, TF-IDF (0.690); word2vec pooling trails at 0.610. In-domain masked-language-model pretraining yields a +0.9-point macro-F1 change over the BERT baseline, which falls inside the binomial noise envelope of a single seed and is therefore inconclusive. Few-shot prompting of the frozen 3-billion-parameter model recovers only about half of supervised performance and collapses onto a handful of topical priors at zero shot. The dominant theme is that gradient-based adaptation, even a 0.5%-parameter adapter, is worth far more than representation cleverness or scale alone. All experiments are seed-pinned and tracked in Weights & Biases.

Index Terms—text classification, word embeddings, TF-IDF, BERT, masked language modelling, in-context learning, QLoRA, parameter-efficient fine-tuning, 20 Newsgroups.

I. INTRODUCTION

Document topic classification is a long-standing benchmark task whose “right answer” has shifted with every wave of NLP methodology, from sparse bag-of-words vectors to dense word embeddings, contextual encoders, and now prompted or adapter-tuned foundation models. The 20 Newsgroups corpus [1] is a convenient testbed for tracing that trajectory: it is small enough to train on a laptop, multi-class enough (20 categories) to expose representational weaknesses, and noisy enough (Usenet quoting, signatures, cross-posting) to reward careful preprocessing.

This work runs a controlled progression of five experiments on a single dataset and evaluation protocol, so that each modelling jump can be attributed to one change rather than a confound. The arc, summarised in Fig. 1, is:

Experiment tracking (Weights & Biases): <https://wandb.ai/danwwaititu-hochschule-luzern/hslu-nalapro>. Source code and this report (report/main.pdf) are available in the accompanying repository. Generative AI usage is disclosed in Section XV.

- 1) **A fixed two-layer MLP** fed three classical input representations: 1-epoch vs. 20-epoch word2vec mean-pools, TF-IDF, and a self-designed $\text{mean} \oplus \text{max}$ pooling variant (Q1).
- 2) **Supervised fine-tuning of BERT-base** with a hyperparameter and ablation sweep, compared head-to-head with Q1 (Q2).
- 3) **Domain-adaptive masked-language-model (MLM) pretraining** on the in-domain corpus, followed by classification fine-tuning (Q3).
- 4) **Zero- and few-shot prompting** of a frozen Llama-3.2-3B instruction model (Q4).
- 5) **QLoRA fine-tuning** of the same Llama model for sequence classification, as a bonus connecting the prompting and supervised worlds.

Contributions. (i) A like-for-like comparison of seven classifiers under one seed, split, and metric, isolating the effect of representation, supervision, and scale. (ii) Evidence that a 0.5%-parameter QLoRA adapter on a 3B decoder surpasses full fine-tuning of a 110M encoder, while zero-/few-shot prompting of the same decoder lags far behind. (iii) A reproducibility-minded reading of small effect sizes: several single-seed deltas reported here, including the Q3 MLM gain, fall within the binomial noise envelope and should not be over-interpreted.

II. BACKGROUND AND RELATED WORK

Classical representations. Bag-of-words with TF-IDF weighting [2], [3] remains a strong, interpretable baseline: it discards word order but preserves every term identity as a sparse coordinate. Distributed word embeddings [4], [5], [6] instead place words in a dense continuous space where geometric proximity encodes distributional similarity; a document is typically reduced to a fixed-size vector by pooling its token vectors, which trades term identity for semantic smoothing.

Contextual encoders. The Transformer [7] and BERT [8] replaced static embeddings with deep context-dependent representations pretrained by masked language modelling. Continuing the MLM objective on in-domain unlabelled text before fine-tuning, known as domain-adaptive pretraining [9], can recover gains when the target domain diverges from the pretraining corpus.

Foundation models and adaptation. Large decoder-only models such as the Llama-3 family [10] can classify by

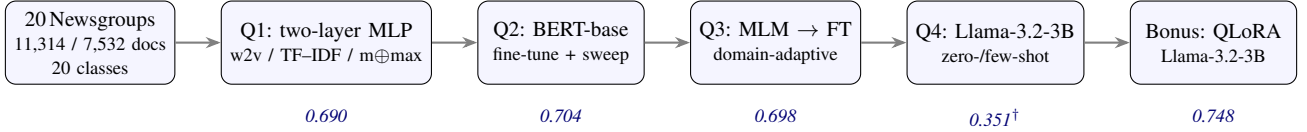


Fig. 1: The five-experiment progression on a single dataset and protocol. Italic numbers are best test-set macro-F1 per stage on the full 7,532-document test set; [†]Q4 is evaluated on a 200-document stratified subset (best few-shot setting) and is not directly comparable.

TABLE I: 20 Newsgroups after preprocessing. Document lengths are token counts after stopword removal and the ≥ 3 -character filter.

Property	Value
Training documents	11,314
Test documents	7,532
Classes	20
Documents per class	≈ 400 –600
Vocabulary (classifier preprocessing)	73,118
word2vec vocabulary (<i>min_count</i> =5)	19,306
Doc. length: median / mean	42 / 96
Doc. length: p95 / max	282 / 6,735

in-context learning [11], predicting a label from a natural-language prompt with zero or a few demonstrations and no weight updates. When supervision is available, parameter-efficient fine-tuning is attractive: LoRA [12] trains small low-rank adapters while freezing the base weights, and QLoRA [13] pushes this further by back-propagating into the adapters through a 4-bit NF4-quantised [14] frozen model, making single-GPU fine-tuning of multi-billion-parameter models feasible.

III. DATASET AND PREPROCESSING

We use the scikit-learn distribution of 20 Newsgroups [15] with its canonical train/test split (11,314 / 7,532 documents, Table I). The dataset is never committed to the repository; it is fetched at runtime. Headers, footers, and quoted text are stripped at load time: these fields leak the label almost perfectly (the originating newsgroup often appears verbatim in the header), so retaining them would inflate every score and invalidate the comparison.

Two preprocessing regimes are used deliberately. For the *classifier* inputs (TF-IDF and the word2vec pooling for the MLP), text is lowercased, tokenised, short tokens (< 3 characters) are dropped, and English stopwords are removed. For *word2vec training*, stopwords are *kept*: skip-gram learns better embeddings when frequent function words remain in the context window, because they carry positional and syntactic co-occurrence signal. The class distribution is mildly imbalanced (Fig. 2a), so accuracy and macro-F1 track each other closely; we report both throughout and treat macro-F1 as primary because it weights every newsgroup equally. Document lengths are heavily right-skewed (Fig. 2b): a median of 42 tokens but a long tail past 6,000, which later motivates a truncation length for the Transformer models.

A stratified 90/10 train/validation split (seed 42) is carved from the training set and reused identically across Q1, Q2,

and Q3, so that validation-driven early stopping and model selection compare like with like. The held-out test set is touched only for final evaluation.

IV. COMMON EXPERIMENTAL SETUP

All randomness derives from a single constant (SEED=42), which seeds Python, NumPy, PyTorch [16], and the hash seed; word2vec receives the same seed and is pinned to a single worker, because gensim’s multi-threaded training is otherwise non-deterministic and would make the 1-epoch vs. 20-epoch comparison meaningless. Every run logs configuration and per-epoch metrics to Weights & Biases. Metrics are computed by one shared function (accuracy, macro-F1, per-class F1, confusion matrix) over the full 20-label range so that classes absent from the predictions still occupy a slot.

Hardware spans three tiers, reflecting model size: the Q1 MLP trains on CPU; the BERT experiments (Q2, Q3) run on Apple-Silicon MPS with fp16/bf16 and pinned memory disabled for numerical safety; the Llama experiments (Q4, bonus) run on CUDA (an RTX 3060 and a Colab A100). The software stack is PyTorch [16], HuggingFace Transformers [17] and PEFT [18], gensim [19], scikit-learn [15], and bitsandbytes for 4-bit quantisation.

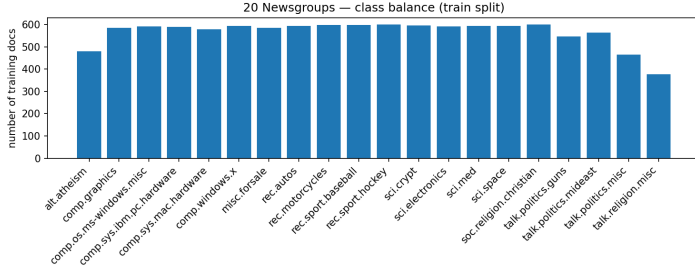
V. EXPERIMENT 1: A FIXED MLP WITH CLASSICAL REPRESENTATIONS

A. Model

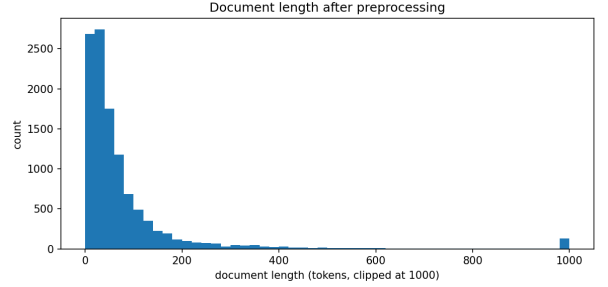
The network is fixed by the specification: a two-layer perceptron, $\text{Linear}(d \rightarrow 256) \rightarrow \text{ReLU} [20] \rightarrow \text{Dropout}(0.3) \rightarrow \text{Linear}(256 \rightarrow 20)$. Dropout sits between the linear layers and is regularisation, not an architectural change; without it the 256-hidden model overfits the 20k-dimensional TF-IDF input badly. Training uses Adam [21] (learning rate 10^{-3} , weight decay 10^{-5}), batch size 64, up to 50 epochs with early stopping (patience 5 on validation loss) and best-epoch weight restoration. Only the input representation, and hence the input dimension d , changes across Q1b–Q1d.

B. Q1b: word2vec embeddings and the effect of training

A skip-gram word2vec model (100-dimensional vectors, window 5, *min_count* = 5) is trained on the stopword-retaining tokens. To answer the specification’s question about the effect of training, we train two checkpoints from byte-identical initial weights: one for a single epoch and one for twenty. The vocabulary is identical in both (19,306 types), so any change is in the geometry, not the support. Fig. 3 projects the 500 most frequent tokens with t-SNE [22]. After one epoch



(a) Documents per class (training split).



(b) Document-length distribution (tokens).

Fig. 2: Dataset structure. Mild class imbalance and a long-tailed length distribution.

the cloud is essentially undifferentiated; after twenty epochs coherent semantic neighbourhoods have formed (vehicle terms cluster apart from religion terms, and so on). Training does not move words randomly: it pulls distributionally similar words together, which is exactly the structure a downstream classifier can exploit. Documents are encoded by mean-pooling their in-vocabulary token vectors (out-of-vocabulary tokens skipped; empty documents mapped to the zero vector). The 20-epoch mean-pool reaches 0.632/0.610 (Table II), the floor that the remaining Q1 variants must beat.

C. Q1c: TF-IDF

The TF-IDF representation uses the 20,000 most frequent unigrams with sublinear term-frequency scaling and a minimum document frequency of two. It reaches 0.696/0.690, beating the word2vec mean-pool by 6.4 accuracy and 8.0 macro-F1 points. The reason is representational: mean-pooling averages a document into a single 100-dimensional point and destroys term identity, whereas TF-IDF keeps every discriminative word as its own coordinate. For topic classification, where the presence of a rare word such as “encryption” or “crucifixion” is nearly decisive, preserving identity beats smoothing similarity. Dense embeddings help most when training data is scarce or when word order matters, neither of which favours mean-pooling here.

D. Q1d: a self-designed experiment (mean $\oplus$ max pooling)

The required “one more experiment that could improve results while the network stays the same” targets the information bottleneck identified in Q1c: pooling. Instead of the mean alone, we concatenate the element-wise mean and the element-wise max of the token vectors, doubling the input to 200 dimensions while leaving the architecture untouched. The intuition is that the max captures salient “peak” features (a strongly category-indicative word) that the mean washes out. In practice it does not help: Q1d scores 0.620/0.600, a slight regression from the mean-only Q1b. The extra dimensions add parameters only to the first linear layer and, on a 100-dimensional embedding space already lossy from averaging, the max channel mostly injects noise rather than signal. The honest reading is that the bottleneck is the embedding space itself, not the pooling operator; richer pooling cannot recover identity information that mean-pooling never had.

TABLE II: Q1 results on the full test set. The fixed MLP, three input representations.

Variant	Representation	Acc.	Macro-F1
Q1b	word2vec mean-pool (100-d)	0.632	0.610
Q1c	TF-IDF (20k unigrams)	0.696	0.690
Q1d	word2vec mean $\oplus$ max (200-d)	0.620	0.600

VI. EXPERIMENT 2: FINE-TUNING BERT

A. Setup and the truncation-length choice

We fine-tune `bert-base-uncased` through the Hugging-Face Trainer [17]. Because attention is quadratic in sequence length and the corpus is long-tailed, the maximum sequence length is a real trade-off. The token-length histogram (Fig. 4) shows a 90th percentile near 256 tokens; we set `max_length` = 256, which covers most documents while remaining sweepable on MPS within an afternoon, and dynamically pad each batch to its longest member rather than to 256.

B. Baseline and sweep

The Q2b baseline (learning rate 2×10^{-5} , three epochs, batch size 16, weight decay 0.01, 10% warmup) reaches 0.708/0.689 on the test set, with a best validation macro-F1 of 0.745. Q2c then sweeps learning rate, sequence length, casing, and an encoder-freeze ablation (Table III, Fig. 5). The best configuration is the highest swept learning rate, 5×10^{-5} , at 0.719/0.704.

Two findings stand out. First, the *frozen-encoder linear probe* (only the classification head trains) collapses to 0.310/0.258: a frozen `bert-base` sentence representation is, on its own, a poor 20-way topic feature, and almost all of BERT’s value here comes from adapting the encoder, not from its off-the-shelf features. Second, several nominally identical configurations differ by up to a point of macro-F1 (e.g. the two 5×10^{-5} , length-256 runs at 0.704 and 0.692; the 2×10^{-5} run at 0.681 vs. Q2b’s 0.689). These gaps are run-to-run variance from non-deterministic MPS kernels, and they set a useful sanity threshold: a single-seed difference below roughly one point should not be read as a real effect, a caution that returns in Q3.

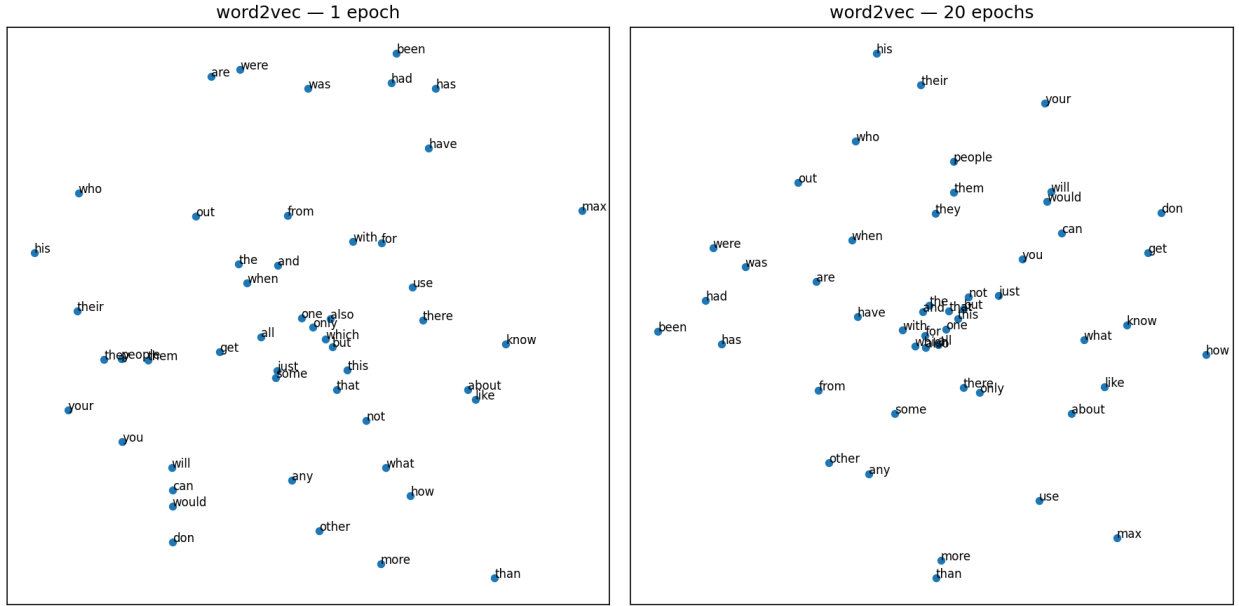


Fig. 3: t-SNE projection of the 50 most frequent word2vec vectors, trained for 1 epoch (left) vs. 20 epochs (right) from identical initial weights. Training forms semantic clusters that are absent after a single epoch.

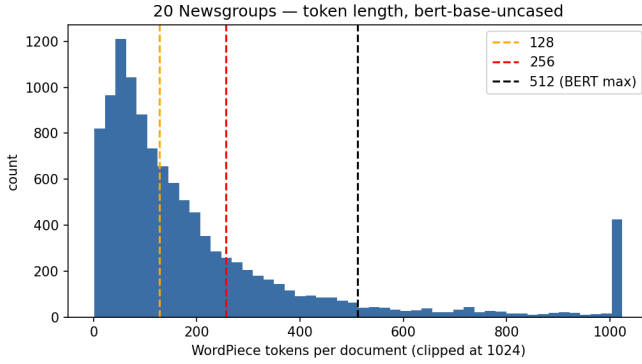


Fig. 4: BERT subword-token lengths over the training corpus (clipped at 1024). The chosen *max_length* = 256 covers roughly the 90th percentile.

TABLE III: Q2c sweep, sorted by macro-F1. All variants fine-tune the full encoder except the linear probe. *m1*: max length.

Variant	LR	m1	Cased	Acc.	Macro-F1
LR 5×10^{-5}	5e-5	256	no	0.719	0.704
LR 3×10^{-5}	3e-5	256	no	0.712	0.695
seqlen 256	5e-5	256	no	0.709	0.692
cased	5e-5	256	yes	0.704	0.688
LR 2×10^{-5}	2e-5	256	no	0.703	0.681
seqlen 128	5e-5	128	no	0.698	0.677
LR 1×10^{-5}	1e-5	256	no	0.693	0.666
frozen probe	1e-3	256	no	0.310	0.258

C. Q1 vs. Q2

The best BERT model improves on the best Q1 model (TF-IDF) by 2.2 accuracy and 1.4 macro-F1 points. The margin is real but modest, and it comes at a steep cost: *bert-base* has ≈ 110 M parameters and fine-tunes in tens of minutes on a

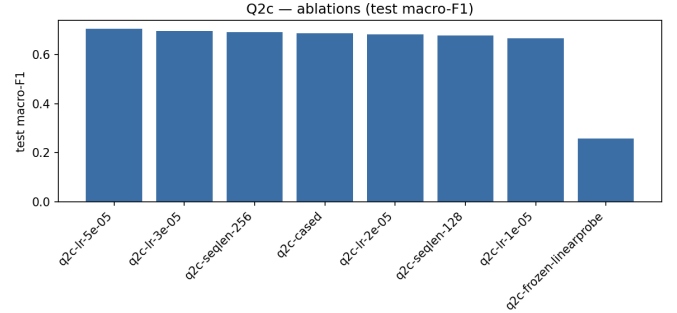


Fig. 5: Test macro-F1 across the Q2c sweep. Fine-tuning the encoder dominates; the frozen linear probe is the floor.

GPU/MPS, against a small MLP head over a sparse vector that trains in seconds on CPU. On this corpus, a well-built TF-IDF baseline is within striking distance of a fine-tuned Transformer, a reminder that classical methods remain competitive when topics are lexically well separated.

VII. EXPERIMENT 3: IN-DOMAIN MLM PRETRAINING, THEN FINE-TUNING

A. Method

Q3 tests the domain-adaptive pretraining hypothesis [9]: that a stage of MLM on the in-domain corpus produces a better encoder than fine-tuning *bert-base-uncased* directly. Stage A continues the MLM objective on the training documents (15% masking, learning rate 5×10^{-5} , three epochs, length 256), with an internal 90/10 split for the MLM eval loss that is independent of the classification validation set to avoid leakage. Stage B is the Q2b fine-tuning recipe applied verbatim, the only difference being that the encoder is

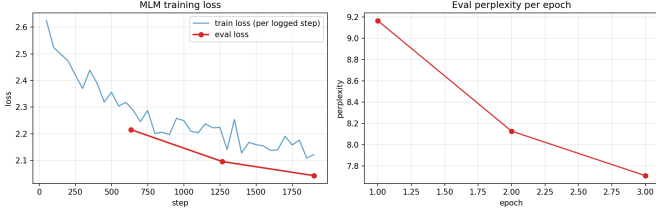


Fig. 6: Stage-A MLM training: loss and perplexity per epoch. Best eval loss 2.04 (perplexity 7.71).

initialised from the MLM checkpoint rather than downloaded fresh. Holding Stage B fixed isolates the encoder-initialisation effect.

B. Results

MLM converges smoothly to a best evaluation loss of 2.04, a perplexity of 7.71 (Fig. 6). The fine-tuned classifier reaches 0.716/0.698, against Q2b’s 0.708/0.689: a change of +0.8 accuracy and +0.9 macro-F1 points. Whether to believe it requires an effect-size check. On 7,532 test documents, the 95% binomial confidence half-width near 71% accuracy is $1.96\sqrt{0.71 \cdot 0.29/7532} \approx 1.0$ point. The observed deltas sit at or just inside that envelope, and they are the same magnitude as the MPS run-to-run variance measured in Q2c. The most defensible conclusion is that in-domain MLM gave at best a marginal, within-noise improvement here. This is consistent with the conditions: the corpus is small ($\sim 10k$ documents), bert-base is already strong on this task, leaving little headroom, and a single seed cannot separate a sub-point effect from chance. Per-class inspection (the side-by-side confusion matrices in the repository) suggests the small gains concentrate on lower-resource, jargon-heavy classes, which is where domain adaptation should help if it helps anywhere.

VIII. EXPERIMENT 4: ZERO- AND FEW-SHOT LLAMA-3

A. Method

Q4 evaluates a frozen Llama-3.2-3B-Instruct [10] loaded in 4-bit NF4 [13], [14]. We treat it as a *generative* classifier, the modern reading of “zero-/few-shot with an instruction model”: the model is prompted with the task, the 20 allowed labels, and (for few-shot) demonstrations, then asked to emit one label, which is parsed back to a class. The alternative of scoring $\log P(\text{label} \mid \text{prompt})$ over the 20 labels is always well-defined but does not test instruction following, the capability actually under evaluation. Decoding is greedy (reproducible), documents are truncated to 1,500 characters, and a parser maps the generation to a label by exact, then substring, then fuzzy (edit-distance) matching, tracking an *invalid rate* for unparseable outputs.

Because a single document costs 2–3 s on the 3060, evaluating all 7,532 test documents three times is impractical; we use a stratified 200-document subset (10 per class, seed 42), reused byte-identically by the bonus experiment so the two are comparable. Few-shot demonstrations are sampled per class from the training set (1 and 3 per class, i.e. 20 and 60 demonstrations), never overlapping the test subset.

TABLE IV: Q4 Llama-3.2-3B on the 200-document subset. Invalid rate is the fraction of unparseable generations.

Setting	Acc.	Macro-F1	Invalid
Zero-shot	0.155	0.132	0.005
Few-shot, $k=1/\text{class}$	0.240	0.246	0.040
Few-shot, $k=3/\text{class}$	0.365	0.351	0.035

B. Results

Results climb steadily with demonstrations (Table IV): zero-shot 0.155/0.132, one-per-class 0.240/0.246, three-per-class 0.365/0.351. The zero-shot failure is structural, not a parsing artefact: the invalid rate is only 0.5%, yet 147 of 200 predictions (73.5%) collapse onto just three labels (talk.religion.misc, talk.politics.misc, talk.politics.guns) and six classes are never predicted at all (Fig. 7). The model emits valid labels but defaults to a few strong topical priors when uncertain. Demonstrations counter this directly: a single example per class is enough to make under-predicted categories (sports, for-sale, hardware) “visible” as legal answers, and three examples begin to teach surface patterns rather than mere label legality, which is why accuracy keeps rising rather than saturating at $k=1$. Even so, the best few-shot setting recovers only about half the macro-F1 of any fine-tuned model: prompting a small frozen model is no substitute for parameter updates on this 20-way task.

IX. BONUS: QLoRA FINE-TUNING OF LLAMA-3

A. Method

The bonus closes the loop by *fine-tuning* the same Llama-3.2-3B for classification under a tight compute budget, using QLoRA [13], [12]. The 4-bit base is frozen and a LlamaForSequenceClassification head is attached; LoRA adapters are injected into the attention projections (q,k,v,o_proj). One detail is decisive: the freshly initialised classification head (score) is placed in modules_to_save so it trains in full precision alongside the adapters. Without this, the random head never updates (LoRA cannot compose into a base module that did not exist at pretraining) and the model silently fails to learn. A two-point sweep is run, $r=16$, $\alpha=32$, $\text{lr}=2 \times 10^{-4}$ versus $r=32$, $\alpha=64$, $\text{lr}=1 \times 10^{-4}$, with the winner chosen by validation macro-F1 ($r=16$ at 0.807 vs. 0.763). The winning adapter trains only $\sim 9.2\text{M}$ parameters, about 0.5% of the model.

B. Results

On the full 7,532-document test set the QLoRA model reaches 0.757/0.748, and on the Q4 200-document subset 0.805/0.799. Against Q4 on the matched subset, this is a jump of 0.65 accuracy over zero-shot and 0.44 over the best few-shot setting: gradient updates dwarf prompting. Against BERT, the comparison is the genuinely interesting one. A 0.5%-parameter adapter on a 3B decoder beats the fully fine-tuned 110M bert-base (Q2c) by 4.4 macro-F1 points on the full test set, despite the decoder-only architecture being a less natural fit for classification than a bidirectional encoder. The

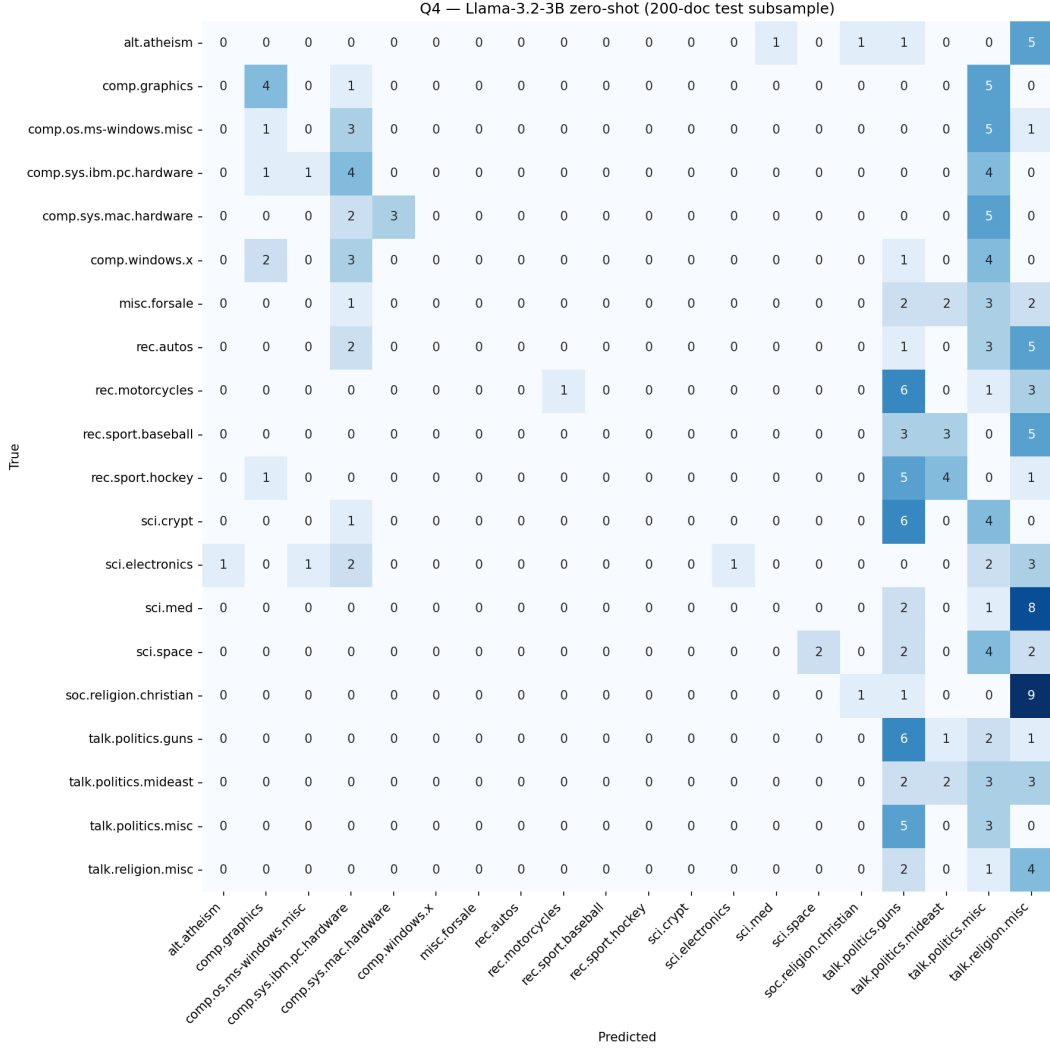


Fig. 7: Zero-shot confusion matrix. Predictions (columns) collapse onto a few politics/religion labels; six newsgroups receive no predictions, which drives the low macro-F1.

extra scale, even accessed through a tiny adapter, more than compensates. The practical appeal is that the result is a single ~50MB adapter file that can be attached to or detached from the shared base model at inference time, at a fraction of the training cost of full fine-tuning.

The per-class view (Fig. 8) ties this back to Q4. The largest F1 gains from fine-tuning fall precisely on the classes zero-shot prompting never predicted (the sports, for-sale, and science groups), while the smallest gains are on the politics/religion labels that zero-shot already over-predicted. Fine-tuning supplies exactly what prompting could not: a calibrated decision boundary over all 20 classes rather than a few topical defaults.

X. CROSS-EXPERIMENT SYNTHESIS

Table V and Fig. 9 collect every result. Three patterns emerge.

Adaptation beats representation, and scale beats neither without adaptation. The single largest lever is whether the model’s weights are updated for the task. The frozen-encoder

probe (0.258) and zero-shot Llama (0.132) are the two worst results in the study despite resting on a 110M encoder and a 3B decoder respectively. The moment gradients flow, even into a 0.5% adapter, performance leaps. Representation choices (TF-IDF vs. embeddings, mean vs. mean $\oplus$ max) move macro-F1 by single-digit points; adaptation moves it by tens.

Diminishing, sometimes invisible, returns higher up. From TF-IDF (0.690) to tuned BERT (0.704) to MLM-then-fine-tune (0.698), the gains are small and, for Q3, within noise. The headroom on cleanly separated topics is limited, and added machinery (a Transformer, an extra pretraining stage) does not guarantee a measurable return on a small corpus.

The biggest jump is QLoRA, by scale via a cheap adapter. QLoRA’s 0.748 is the only result that clears the BERT plateau by a clear margin, and it does so with fewer trainable parameters than the BERT head sees gradients for. For a practitioner with one GPU, parameter-efficient fine-tuning of a small foundation model is the most cost-effective way to specialise for this task; prompting the same model without fine-tuning is the least.

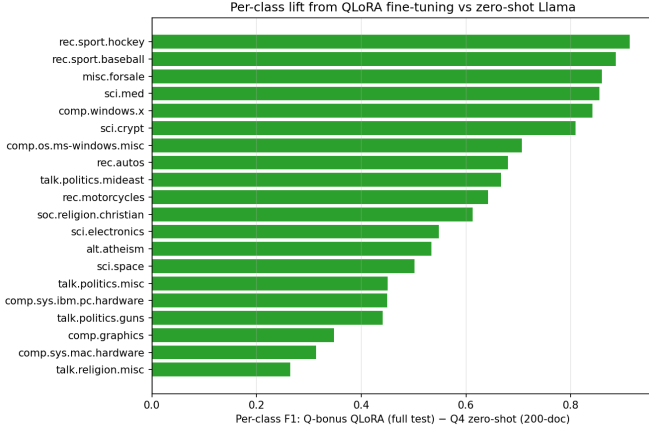


Fig. 8: Per-class F1 lift of QLoRA fine-tuning over zero-shot Llama, sorted descending. Gains concentrate on the classes zero-shot ignored. (QLoRA on the full test set vs. zero-shot on the 200-doc subset; directional, since the splits differ.)

TABLE V: Master results. Full test set unless marked [†] (200-doc subset). Best full-test macro-F1 in bold.

Exp.	Method	Acc.	Macro-F1
Q1b	MLP + word2vec mean	0.632	0.610
Q1c	MLP + TF-IDF	0.696	0.690
Q1d	MLP + word2vec mean $\oplus$ max	0.620	0.600
Q2b	BERT-base (baseline)	0.708	0.689
Q2c	BERT-base (tuned)	0.719	0.704
Q3	MLM $\rightarrow$ fine-tune	0.716	0.698
B	QLoRA Llama-3.2-3B	0.757	0.748
Q4	Llama zero-shot [†]	0.155	0.132
Q4	Llama few-shot $k=3$ [†]	0.365	0.351
B	QLoRA Llama [†]	0.805	0.799

XI. CHALLENGES FACED

Label leakage. The single highest-impact decision was stripping headers, footers, and quotes. Early exploratory runs that retained them produced implausibly high accuracy; the originating newsgroup leaks through metadata, and honest evaluation required removing it before any modelling.

Non-determinism on Apple Silicon. BERT fine-tuning on MPS is not bitwise reproducible even with a fixed seed, producing the ~ 1 -point spread seen in Table III. This complicated the Q3 conclusion and forced an explicit effect-size argument rather than a naive “higher number wins”.

Compute for the LLM stages. Generative classification at 2–3 s per document made full-test evaluation of Q4 infeasible, motivating the 200-document subset. QLoRA was likewise constrained to a two-point sweep on a single A100 session rather than a broad search.

The silent QLoRA-head failure. The most instructive bug was the classification head not training under LoRA. Loss decreased while validation metrics stayed at chance, because the adapters were learning around a frozen, randomly initialised head. Placing `score` in `modules_to_save` fixed it; the symptom (training loss falling, metrics flat) is worth recognising.

Generative output parsing. The model occasionally returned a label with extra words or formatting. A layered parser (exact, then substring, then edit-distance) with an explicit invalid-rate diagnostic kept parsing from being confused with genuine misclassification, which was essential to correctly diagnosing the zero-shot class-bias failure.

XII. LESSONS LEARNT

A small number of transferable lessons recur. First, *measure the noise floor before trusting a delta*: the binomial confidence interval and the observed MPS variance were both about one point, which reframed the Q3 “improvement” as inconclusive. Second, *a strong classical baseline is a cheap insurance policy*: TF-IDF came within 1.4 macro-F1 points of fine-tuned BERT for a tiny fraction of the cost, and any claim that a heavy model is necessary should be checked against it. Third, *adaptation is the dominant variable*; effort spent enabling weight updates (unfreezing an encoder, correctly training a LoRA head) returns far more than effort spent engineering representations. Fourth, *centralise determinism and evaluation*: a single seeding entry point and a single metrics function removed a whole class of subtle inconsistencies and made cross-experiment comparison trustworthy. Finally, *keep logic out of notebooks*: putting all reusable code in a tested package, with notebooks as thin orchestrators, made the five experiments reproducible and the results easy to re-derive.

XIII. LIMITATIONS

The headline limitation is single-seed reporting. A proper variance estimate needs 3–5 seeds per condition; without it, sub-point differences (Q3 vs. Q2b, and within the Q2c sweep) cannot be called significant, and we have deliberately refrained from doing so. Q4 and the bonus subset evaluation use 200 documents, which has wide per-class confidence intervals (10 test documents per class), so those numbers indicate trends rather than precise levels. The Transformer models truncate at 256 tokens and the LLMs at 1,500 characters, discarding the long tail of the length distribution. Q3 reuses Q2b’s hyperparameters rather than sweeping its own, so its absolute number may understate what domain-adaptive pretraining can achieve. Finally, comparisons across model classes mix evaluation splits (full test vs. subset) and architectures (encoder vs. decoder), so the cross-paradigm rankings are directional rather than exact.

XIV. FUTURE WORK

Several extensions follow directly. Repeating every experiment over multiple seeds would turn the marginal Q3 result into a properly powered test of domain-adaptive pretraining and put error bars on the cross-experiment plot. A dedicated hyperparameter sweep for Q3, and a larger QLoRA sweep over rank, target modules, and learning rate, would establish whether the gaps observed here are robust. Evaluating Q4 and the QLoRA model on the full test set (given more compute) would remove the subset caveat. Methodologically, task-adaptive pretraining and longer-context truncation are natural follow-ups for the BERT line, and instruction-style fine-tuning

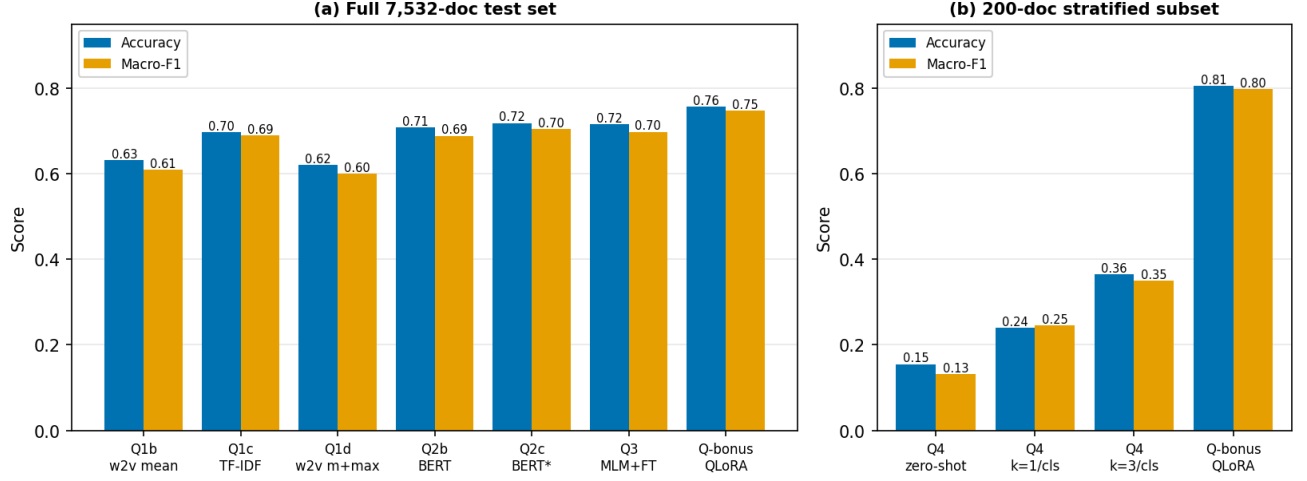


Fig. 9: Cross-experiment summary. (a) Full 7,532-document test set; (b) the matched 200-document subset, isolating prompting vs. QLoRA fine-tuning of the same Llama model. Subset and full-test numbers are not directly comparable.

or calibrated label scoring could lift the prompting baseline. Finally, a class-weighted or focal loss might address the persistent weakness on the low-resource talk.* newsgroups visible in every confusion matrix.

XV. CONCLUSION

Across five experiments on 20 Newsgroups under one protocol, the determining factor for classification quality was not representation sophistication or raw model scale but whether the model’s weights were adapted to the task. A fixed MLP showed that term-identity-preserving TF-IDF beats lossy embedding pooling; fine-tuned BERT edged past it at much higher cost; in-domain MLM pretraining added only a within-noise change; zero-/few-shot prompting of a 3B model lagged far behind; and a 0.5%-parameter QLoRA adapter on that same model delivered the best result of the study. Parameter-efficient fine-tuning of a small foundation model emerges as the most cost-effective path to strong performance, and rigorous effect-size reasoning emerges as the discipline that keeps such a comparison honest.

DECLARATIONS

Data availability. 20 Newsgroups is public and fetched at runtime via scikit-learn; it is not redistributed in the repository.

Code and experiment tracking. All code, notebooks, and this report (report/main.pdf) are in the accompanying repository. Runs are logged to Weights & Biases: <https://wandb.ai/danwwaititu-hochschule-luzern/hslu-nalapro>.

Tools used. Python with PyTorch, HuggingFace Transformers and PEFT, bitsandbytes, gensim, scikit-learn, NLTK, matplotlib/seaborn, Weights & Biases, and uv for environment management. Models: bert-base-uncased and meta-llama/Llama-3.2-3B-Instruct.

Generative AI disclosure. Anthropic’s Claude (via Claude Code) was used to assist with code scaffolding, refactoring, documentation, figure generation, and drafting of this report. All experimental design, results, and analysis were reviewed

and verified by the author, who understands and can explain the submitted code and findings.

Author contributions. Dan Waititu (sole author): conception, implementation, analysis, and writing.

Funding and conflicts. No external funding; no competing interests. The work involves no human subjects or sensitive data.

REFERENCES

- [1] K. Lang, “Newsweeder: Learning to filter netnews,” in *Proceedings of the Twelfth International Conference on Machine Learning (ICML)*. Morgan Kaufmann, 1995, pp. 331–339.
- [2] K. Spärck Jones, “A statistical interpretation of term specificity and its application in retrieval,” *Journal of Documentation*, vol. 28, no. 1, pp. 11–21, 1972.
- [3] G. Salton and C. Buckley, “Term-weighting approaches in automatic text retrieval,” *Information Processing & Management*, vol. 24, no. 5, pp. 513–523, 1988.
- [4] T. Mikolov, K. Chen, G. Corrado, and J. Dean, “Efficient estimation of word representations in vector space,” in *Workshop Track Proceedings of the International Conference on Learning Representations (ICLR)*, 2013, arXiv:1301.3781.
- [5] T. Mikolov, I. Sutskever, K. Chen, G. S. Corrado, and J. Dean, “Distributed representations of words and phrases and their compositionality,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 26, 2013.
- [6] J. Pennington, R. Socher, and C. D. Manning, “GloVe: Global vectors for word representation,” in *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2014, pp. 1532–1543.
- [7] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.
- [8] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*, 2019, pp. 4171–4186.
- [9] S. Gururangan, A. Marasović, S. Swayamdipta, K. Lo, I. Beltagy, D. Downey, and N. A. Smith, “Don’t stop pretraining: Adapt language models to domains and tasks,” in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020, pp. 8342–8360.
- [10] A. Grattafiori et al., “The Llama 3 herd of models,” *arXiv preprint arXiv:2407.21783*, 2024.

- [11] T. B. Brown *et al.*, “Language models are few-shot learners,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 1877–1901.
- [12] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “LoRA: Low-rank adaptation of large language models,” in *International Conference on Learning Representations (ICLR)*, 2022, arXiv:2106.09685.
- [13] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, “QLoRA: Efficient finetuning of quantized LLMs,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023, arXiv:2305.14314.
- [14] T. Dettmers, M. Lewis, Y. Belkada, and L. Zettlemoyer, “LLM.int8(): 8-bit matrix multiplication for transformers at scale,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, 2022, arXiv:2208.07339.
- [15] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [16] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga *et al.*, “PyTorch: An imperative style, high-performance deep learning library,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019.
- [17] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz *et al.*, “Transformers: State-of-the-art natural language processing,” in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations (EMNLP)*, 2020, pp. 38–45.
- [18] S. Mangrulkar, S. Gugger, L. Debut, Y. Belkada, S. Paul, and B. Bossan, “PEFT: State-of-the-art parameter-efficient fine-tuning methods,” <https://github.com/huggingface/peft>, 2022.
- [19] R. Rehůřek and P. Sojka, “Software framework for topic modelling with large corpora,” in *Proceedings of the LREC 2010 Workshop on New Challenges for NLP Frameworks*, 2010, pp. 45–50.
- [20] V. Nair and G. E. Hinton, “Rectified linear units improve restricted boltzmann machines,” in *Proceedings of the 27th International Conference on Machine Learning (ICML)*, 2010, pp. 807–814.
- [21] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in *International Conference on Learning Representations (ICLR)*, 2015, arXiv:1412.6980.
- [22] L. van der Maaten and G. Hinton, “Visualizing data using t-SNE,” *Journal of Machine Learning Research*, vol. 9, pp. 2579–2605, 2008.